DETAIL SPECIFICATION SHEET

EXPANSION JOINTS, PIPE, METALLIC, CORRUGATED BELLOWS, EXTERNALLY PRESSURIZED

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17813.

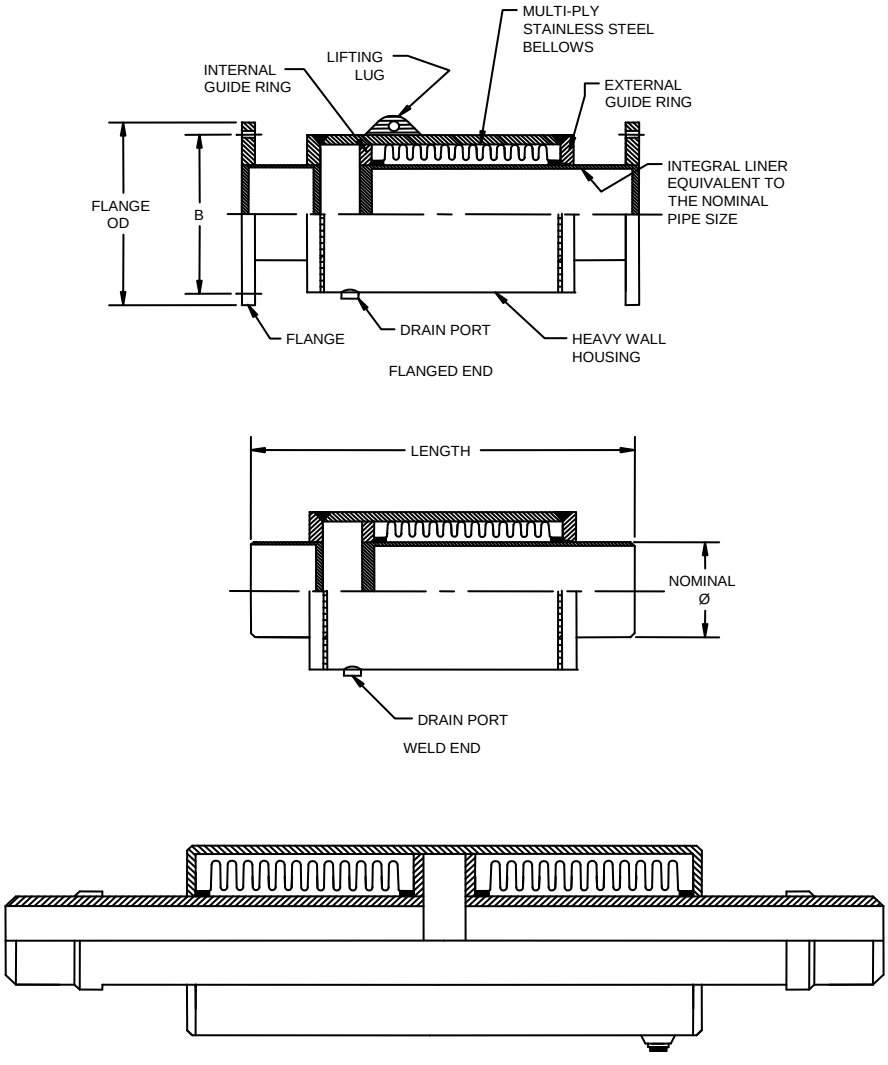


FIGURE 1. Expansion joint, externally pressurized.

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DN	C-NPS inch (mm)	Flange OD inch (mm) 1/	B (bolt circle diameter) inch (mm) 1/
50	2 (50.8)	6 (152.4)	4.75 (120.7)
65	2.5 (76.2)	7 (177.8)	5.5 (139.7)
80	3 (76.2)	7.5 (190.5)	6 (152.4)
90	3.5 (101.6)	8.5 (215.9)	7 (177.8)
100	4 (101.6)	9 (228.6)	7.5 (190.5)
125	5 (127)	10 (254.0)	8.5 (215.9)
150	6 (152.4)	11 (279.4)	9.5 (241.3)
200	8 (203.2)	13.5 (342.9)	11.75 (298.5)
250	10 (254)	16 (406.4)	14.25 (362.0)
300	12 (304.8)	19 (482.6)	17 (431.8)
350	14 (355.6)	21 (533.4)	18.75 (476.3)
400	16 (406.4)	23.5 (596.9)	21.25 (539.8)
450	18 (457.2)	25 (635)	22.75 (577.9)
500	20 (508)	27.5 (698.5)	25 (635)
550	22 (558.8)	-	-
600	24 (609.6)	32 (812.8)	29.5 (749.3)

1/ See ASME B16.5, flange dimensions,150 pound.

NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.
2. The illustration is for identification purposes only and is not intended to restrict designs and shapes of fastening devices, which conform to the requirements and are within the envelope dimensions.

FIGURE 1. Expansion joint, externally pressurized – Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Construction: Externally pressurized expansion joints shall be constructed of welded metal piping. Each unit shall be constructed with both internal and external guide rings. Each unit shall be equipped with a drain port to be located on the outer shell of the unit. The design shall incorporate a totally enclosed, externally pressurized stainless steel bellows that is protected from by a heavy-walled shroud designed for full line pressure. The bellows shall be of single-ply, or multi-ply construction.

The classification of the expansion joints covered under this specification sheet are as follows:

Type I - shall have corrugated bellows.

Class 4 - shall be of the externally pressurized variety either single or double bellows.

The bellows in class 4 expansion joints shall be designed and manufactured in accordance with the EJMA Standards. Sizes up to and including 10 inches shall be formed from seamless tubing

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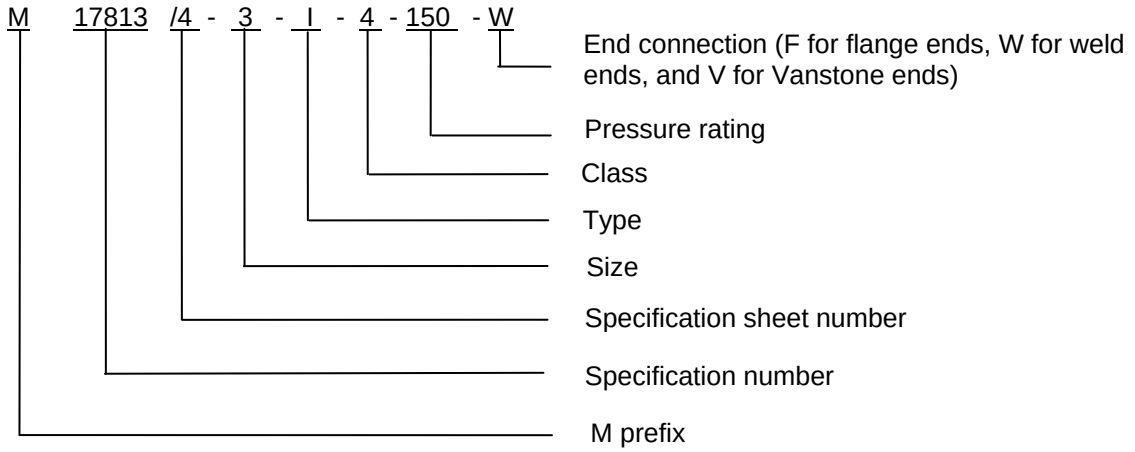
or from a tube having only one longitudinal welded seam. For sizes larger than 10 inches, more than one longitudinal seam is permissible. In no case will circumferential seams be accepted unless the requirements of a particular application require such construction and the design is approved by the agency concerned. Convolutions shall be symmetrical and of similar dimensions.

Single or double bellows construction shall be specified in MIL-DTL-17813.

Materials and finishes.

Materials and finishes for the bellows and flanges shall be in accordance with MIL-DTL-17813.

The Part or Identifying Number (PIN) consists of the letters M, the basic specification number, specification sheet number, size, type, class, pressure rating, and end connection.



PIN example: M17813/4-3-I-4-150-W

Referenced documents. In addition to MIL-DTL-17813, this document references the following:

ASME B16.5
EJMA Standards

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2009-102)

Review activities:
Army - CE
Navy – MC, SA
Air Force - 71

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